



小美技术 工业通信与数字化解决方案

聚焦制造业现场连接、数据采集与系统集成，为企业提供稳定、安全、可持续演进的技术底座。

[进入文档中心](#)[查看 API 示例](#)

稳定可靠

采用工程化交付流程与标准化组件，保障项目从 PoC 到规模化落地的一致性。

行业实践

覆盖 PLC、边缘网关、协议转换与云端集成，面向工业现场真实业务场景。

安全合规

支持权限分层、访问审计与数据传输加密，满足企业级治理与运维要求。

服务能力

- 工业协议接入：Modbus、PROFINET、PROFIBUS、EtherCAT 等。
 - 数据平台建设：采集、清洗、建模与可视化闭环。
 - 系统集成实施：MES、ERP、SCADA 的接口联动与流程打通。
-

交付流程

1. 需求澄清与现场评估
2. 方案设计与分阶段实施
3. 联调验收与运维陪跑

Runtime API Examples

This page demonstrates usage of some of the runtime APIs provided by VitePress.

The main `useData()` API can be used to access site, theme, and page data for the current page. It works in both `.md` and `.vue` files:

```
<script setup>
import { useData } from 'vitepress'

const { theme, page, frontmatter } = useData()
</script>
```

Results

Theme Data

```
<pre>{{ theme }}</pre>
```

Page Data

```
<pre>{{ page }}</pre>
```

Page Frontmatter

```
<pre>{{ frontmatter }}</pre>
```

md

Results

Theme Data

```
{
  "lastUpdated": {
    "text": "最后更新",
    "formatOptions": {
      "dateStyle": "full",
      "timeStyle": "medium"
    }
  },
  "search": {
    "provider": "local",
    "options": {
      "translations": {
        "button": {
          "buttonText": "搜索文档",
          "buttonAriaLabel": "搜索文档"
        },
        "modal": {
          "noResultsText": "无法找到相关结果",
          "resetButtonText": "清除查询条件",
          "footer": {
            "selectText": "选择",
            "navigateText": "切换",
            "closeText": "关闭"
          }
        }
      }
    }
  },
  "siteTitle": "小美技术",
  "logo": {
    "src": "/logo/xm-logo.png",
    "alt": "小美技术 Logo"
  },
  "nav": [
    {
      "text": "首页",
      "link": "/"
    }
  ]
}
```

```

    "text": "远程IO系统",
    "link": "/remoteIO/"
  },
  {
    "text": "飞达控制器",
    "link": "/feeder-controller/"
  },
  {
    "text": "定制项目",
    "link": "/custom/"
  },
  {
    "text": "关于与支持",
    "items": [
      {
        "text": "PDF手册",
        "link": "/pdf/"
      },
      {
        "text": "Markdown 示例",
        "link": "/web/markdown-examples"
      },
      {
        "text": "Runtime API 示例",
        "link": "/web/api-examples"
      },
      {
        "text": "本地开发运行",
        "link": "/web/localsetup"
      },
      {
        "text": "波形图示例",
        "link": "/web/wavedrom"
      }
    ]
  },
  {
    "text": "下载中心",
    "link": "https://github.com/XiaomeiTech/XiaomeiTech.github.io/releases"
  }
],
"sidebar": {
  "/remoteIO/": [
    {

```

```

"text": "远程IO系统",
"items": [
  {
    "text": "系统概述",
    "link": "/remoteIO/"
  }
],
},
{
"text": "耦合器",
"items": [
  {
    "text": "EtherCAT耦合器",
    "link": "/remoteIO/EtherCAT/"
  },
  {
    "text": "PROFINET耦合器",
    "link": "/remoteIO/PROFINET/"
  },
  {
    "text": "EtherNet/IP耦合器",
    "link": "/remoteIO/EtherNetIP/"
  },
  {
    "text": "Modbus耦合器",
    "link": "/remoteIO/Modbus/"
  }
],
},
{
"text": "拓展模块",
"items": [
  {
    "text": "模拟量输入AI",
    "link": "/remoteIO/AI/"
  },
  {
    "text": "模拟量输出AO",
    "link": "/remoteIO/AO/"
  },
  {
    "text": "数字量输入DI",
    "link": "/remoteIO/DI/"
  },
  },

```

```

    {
      "text": "数字量输出DO",
      "link": "/remoteIO/DO/"
    }
  ]
},
"/feeder-controller/": [
  {
    "text": "飞达控制器",
    "items": [
      {
        "text": "控制器概述",
        "link": "/feeder-controller/"
      },
      {
        "text": "控制器说明",
        "link": "/feeder-controller/manual/"
      }
    ]
  }
],
"/custom/": [
  {
    "text": "定制项目",
    "items": [
      {
        "text": "项目概览",
        "link": "/custom/"
      }
    ]
  }
],
"/pdf/": [
  {
    "text": "PDF手册",
    "items": [
      {
        "text": "导出说明",
        "link": "/pdf/"
      },
      {
        "text": "数据手册",
        "link": "/pdf/datasheet"
      }
    ]
  }
]

```

```

    },
    {
      "text": "应用指南",
      "link": "/pdf/app-guide"
    },
    {
      "text": "单品手册",
      "link": "/pdf/product-manual"
    },
    {
      "text": "系列合并手册",
      "link": "/pdf/series-manual"
    },
    {
      "text": "法律与合规",
      "link": "/pdf/legal"
    }
  ]
}
],
"/web/": [
  {
    "text": "关于小美技术",
    "items": [
      {
        "text": "Markdown 示例",
        "link": "/web/markdown-examples"
      },
      {
        "text": "Runtime API 示例",
        "link": "/web/api-examples"
      },
      {
        "text": "本地开发运行",
        "link": "/web/localsetup"
      }
    ]
  }
]
}
]
}
}
}

```

Page Data

```
{
  "title": "Runtime API Examples",
  "description": "",
  "frontmatter": {
    "outline": "deep"
  },
  "headers": [],
  "relativePath": "web/api-examples.md",
  "filePath": "web/api-examples.md",
  "lastUpdated": 1777351593000
}
```

Page Frontmatter

```
{
  "outline": "deep"
}
```

More

Check out the documentation for the [full list of runtime APIs](#).

VitePress Local Setup and Usage Guide

This guide explains how to install Node.js, switch Node versions, and run this project locally on Windows.

0. Quick install and usage (Windows)

If you just want the shortest path, run in this order:

1. Install Node.js 20.x LTS from official MSI:

- <https://nodejs.org/en/download>

2. Open a new PowerShell and verify:

```
node -v  
npm -v
```

powershell

3. In project root, install dependencies:

```
npm install
```

powershell

4. Start local docs site:

```
npm run docs:dev
```

powershell

5. Build production docs:

```
npm run docs:build
```

powershell

6. If something is weird, clean and rebuild:

```
npm run rebuild
```

powershell

1. Recommended versions

- Node.js: 20.x LTS
- npm: bundled with Node.js

This repository is validated with Node.js 20 in CI.

2. Install Node.js from official website (MSI)

Use the official Node.js MSI installer first.

1. Open the official Node.js download page:
 - <https://nodejs.org/en/download>
2. Download the Windows Installer (.msi) for Node.js 20.x LTS.
3. Run the MSI installer and keep default options.
4. Restart your terminal.

Verify installation:

```
node -v  
npm -v
```

powershell

3. Install project dependencies

In project root:

```
npm install
```

powershell

4. Start local development server

```
npm run docs:dev
```

powershell

After startup, open the local URL shown in terminal (usually `http://localhost:5173`).

5. Build and preview

Build docs:

```
npm run docs:build
```

powershell

Preview built site:

```
npm run docs:preview
```

powershell

6. Clean cache and rebuild

Only clean cache:

```
npm run clean:cache
```

powershell

Fast rebuild (clean + build):

```
npm run rebuild:fast
```

powershell

Full rebuild (clean + fresh install + build):

```
npm run rebuild
```

powershell

6.1 Common usage commands

Start development server:

```
npm run docs:dev
```

powershell

Build static site:

```
npm run docs:build
```

powershell

Export PDFs (same pipeline as GitHub Actions):

```
npm run docs:pdf
```

powershell

Preview build output:

```
npm run docs:preview
```

powershell

Clean cache only:

```
npm run clean:cache
```

powershell

Fast rebuild:

```
npm run rebuild:fast
```

powershell

6.2 PDF Export Pipeline

This project uses `vitepress-export-pdf` (Puppeteer-based) to convert Markdown pages to PDF.

Export all PDFs

```
npm run docs:build  
npm run docs:pdf
```

powershell

PDFs are written to `artifacts/pdf/` :

File	Config	Content
<code>datasheet.pdf</code>	<code>pdf-configs/datasheet.mts</code>	Product spec overview
<code>app-guide.pdf</code>	<code>pdf-configs/app-guide.mts</code>	Dev setup & API examples
<code>product-manual.pdf</code>	<code>pdf-configs/product-manual.mts</code>	Single product full manual
<code>series-manual.pdf</code>	<code>pdf-configs/series-manual.mts</code>	All products combined

Customize PDF export

Edit the corresponding file in `pdf-configs/` to adjust:

- `routePatterns` — which pages to include
- `pdfOptions` — paper format, margins, header/footer
- `outFile` — output filename

CI / GitHub Actions

The workflow `.github/workflows/markdown-pdf.yml` runs on every push to `main` / `master` that touches `markdown/**/*.md`. It builds the site, exports PDFs, and publishes them to [GitHub Releases](#).

7. Optional: switch Node versions with nvm-windows

If you need to switch between multiple Node versions, use nvm-windows.

7.1 Install nvm-windows

1. Download installer from:
 - <https://github.com/coreybutler/nvm-windows/releases>
2. Run installer and restart terminal.

7.2 Install and switch versions

Run in PowerShell:

```
nvm install 20.19.0
nvm use 20.19.0
node -v
npm -v
```

Optional checks:

```
nvm list
nvm current
```

Example: switch to Node 22 for testing, then switch back to 20.

```
nvm install 22.15.0
nvm use 22.15.0
node -v

nvm use 20.19.0
node -v
```

8. Common issues

8.1 node command not found after MSI install

- Restart terminal or restart Windows.
- Reinstall Node.js MSI and ensure Add to PATH is enabled.

8.2 nvm command not found

- Close and reopen terminal.
- Confirm nvm-windows is installed in PATH.

8.3 npm ci fails in rebuild

This usually means package-lock.json is out of sync with package.json.

Fix:

```
npm install  
npm run rebuild
```

8.4 Port already in use

Use another port:

```
npx vitepress dev docs --port 5174
```

Markdown Extension Examples

This page demonstrates some of the built-in markdown extensions provided by VitePress.

Syntax Highlighting

VitePress provides Syntax Highlighting powered by [Shiki](#), with additional features like line-highlighting:

Input

```
```js{4}
export default {
 data () {
 return {
 msg: 'Highlighted!'
 }
 }
}
```
```

md

Output

```
export default {
  data () {
    return {
      msg: 'Highlighted!'
    }
  }
}
```

js

Custom Containers

Input

```
 ::: info
```

```
 This is an info box.
```

```
 :::
```

```
 ::: tip
```

```
 This is a tip.
```

```
 :::
```

```
 ::: warning
```

```
 This is a warning.
```

```
 :::
```

```
 ::: danger
```

```
 This is a dangerous warning.
```

```
 :::
```

```
 ::: details
```

```
 This is a details block.
```

```
 :::
```

Output

INFO

This is an info box.

TIP

This is a tip.

WARNING

This is a warning.

DANGER

This is a dangerous warning.

► Details

More

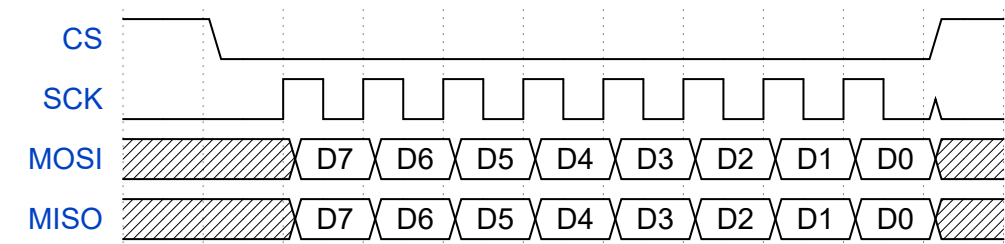
Check out the documentation for the [full list of markdown extensions](#).

绘制 SPI 波形图

在 VitePress 中绘制 WaveDrom 波形，通常可以通过配置 Markdown 插件（例如 `markdown-it-wavedrom` 或 `vitepress-plugin-wavedrom`）来实现。以下是一个标准的 SPI 通信波形示例（SPI Mode 0）。

SPI (Serial Peripheral Interface) - Mode 0

特性：你提供的 Mode 0 经典时序，CS 拉低片选有效，SCK 空闲为低（CPOL=0），在第一个上升沿采样数据（CPHA=0）。



I2C (Inter-Integrated Circuit)

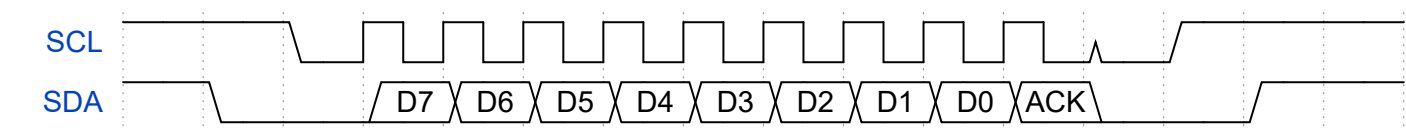
特性：

Start 条件：SCL 为高电平时，SDA 产生下降沿。

数据传输：SCL 高电平时采样，SDA 在 SCL 低电平时允许翻转。

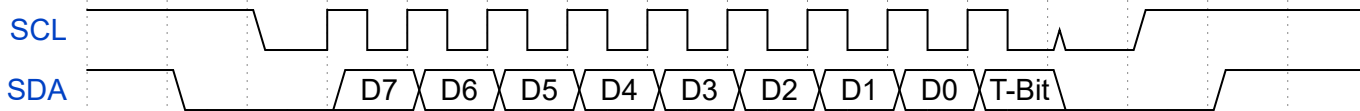
ACK 应答：第 9 个时钟周期，接收方拉低 SDA 作为应答。

Stop 条件：SCL 为高电平时，SDA 产生上升沿。



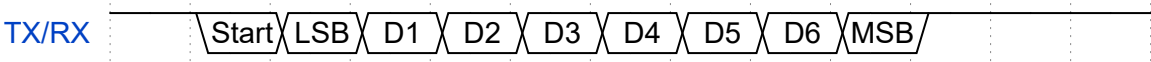
I3C (Improved Inter-Integrated Circuit) - SDR 模式

特性：I3C 向下兼容 I2C，但在高速单数据速率（SDR）模式下，使用推挽（Push-Pull）驱动。与 I2C 最大的不同是，数据传输阶段用 1 位的 T-Bit（Transition Bit，通常用作奇偶校验或结束标志）替代了 I2C 的 ACK 位。



UART (Universal Asynchronous Receiver-Transmitter)

特性：异步通信，无时钟线。默认空闲为高电平（1），通过拉低产生 Start Bit（0），紧接着传输 8 位数据（通常是 LSB 优先，先传 D0），最后通过拉高产生 Stop Bit（1）。



注意事项：

在 VitePress 中实际预览此波形时，依赖于 `vitepress-plugin-wavedrom` 及 `wavedrom` 包提供在网页的渲染支持。现在已经为您配置好。您可以本地运行 `npm run docs:dev` 进行效果预览！